Module 3

# Forwards & Futures

September 10 & 15, 2025

## Forwards

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Contract:

- legal obligation to buy (sell) an asset (e.g., stock, bond, foreign currency, commodity, crypto currency, etc)
    - obligation to buy – "long" the forward
    - obligation to sell – "short" the forward
- at a pre-specific future date (maturity date / expiration date) – T
- in a pre-specified amount (contract size)
- at a pre-specified price (forward price, "strike") – K
- with a pre-agreed delivery method (cash/physical settlement)
- example:

  buy 1000 shares (contract size) of AAPL (asset) on Friday March 20, 2026 (maturity date) at $250.00 (forward price/strike). The share certificates will be delivered to the buyer on Tuesday March 24, 2026 (physical settlement, settlement date)

Valuation:

Assumptions:

- "frictionless market": unlimited cash borrow/lending at risk free rate, unlimited long/short selling with no transaction cost (zero trade commission, bid/ask spread and market impact) and tax

- no counterparty risk: the parties on both side of transaction will always fulfill their obligations in the contract

- asset for investment (security)

- for simplicity assume interest rate r is constant, continuous compounding

Payoff at maturity:

- Long: $F_T = S_T - K$ (agree to buy asset)
- Short: $F_T = K - S_T$ (agree to sell asset)

Price of the contract at inception:

- Choose K such that $F_0 = 0$

i) case of non-dividend paying stock:

→ $K = S_0 e^{rT}$ "fair price"

Consider two portfolios at time t = 0:

portfolio A: one long forward $F_0$ + cash = $Ke^{-rT}$

portfolio B: one unite of asset $S_0$

At t = T, $A_T = F_T + K = S_T = B_T$,

→ at t = 0, $A_0 = B_0$ → $F_0 + Ke^{-rT} = S_0$

$F_0 = 0$ at t = 0 → $K = S_0 e^{rT}$

Arbitrage opportunities if $K \neq S_0 e^{rT}$

- $K > S_0 e^{rT}$ :

  at t = 0, short forward, borrow cash = $S_0$ to buy stock;

  at T, close forward with S, receive K, use $S_0 e^{rT}$ to repay bank,

  → pocket $K - S_0 e^{rT}$

- $K < S_0 e^{rT}$ :

  at t = 0, long forward, short stock at $S_0$, deposit the proceed to earn r;

  at T, close forward with deposit, use the stock got from the forward to close the short stock position,

  → pocket $S_0 e^{rT} - K$

ii) case of stock with known cash dividend schedule:

S pays dividend $D_1, \ldots, D_n$ at time $t_1, \ldots, t_n$ $(0 < t_1, \ldots, t_n < T)$

$$PV(D) = \sum_{i=1}^{n} D_i e^{-rt_i}$$

Consider the above portfolio A, and

portfolio B': one unite of asset $S_0$ and borrowing of PV(D) at rate r.

At t = T, both = $S_T$ → at t = 0, $F_0 + Ke^{-rT} = S_0 - PV(D)$

$F_0 = 0$ at t = 0 → $K = (S_0 - PV(D))\, e^{rT}$

Arbitrage opportunities if $K \neq (S_0 - PV(D))\, e^{rT}$ :

- $K > (S_0 - PV(D))\, e^{rT}$ :

  at t = 0, short forward, taking a series loans with amount $D_1 e^{-rt_1}, ..., D_n e^{-rt_n}, S_0 - PV(D)$ maturing at $t_1$, …, $t_n$ , T, use the sum ( = PV(D) + $S_0$ – PV(D) = $S_0$ ) to purchase stock at $S_0$ .

  At t = T, close forward to receive K ( > $(S_0 - PV(D))\, e^{rT}$ ).

  Use $(S_0 - PV(D))\, e^{rT}$ to repay the loan (loan $D_i e^{-rt_i}$ paid off at $t_i$ with dividend $D_i$)

  → risk free profit = $K - (S_0 - PV(D))\, e^{rT} > 0$

- $K < (S_0 - PV(D))\, e^{rT}$ :

  at t = 0, long forward, short stock, invest the proceeding of short sale in r

  At $t_i$ borrow $D_i$ to cover dividend on the short stock position

  At t = T, total cash in hand = $S_0\, e^{rT} - \sum_{i=1}^{n} D_i e^{r(T-t_i)} = (S_0 - PV(D))\, e^{rT}$

  use K ( < $(S_0 - PV(D))\, e^{rT}$ ) to close the forward and cover the short position in S

  → risk free profit = $(S_0 - PV(D))\, e^{rT} - K > 0$

Note:

- If the yield curve is not constant then PV(D) should be computed as

  $$PV(D) = \sum_{i=1}^{n} D_i e^{-r_i t_i}$$

  where $r_i$ is the borrow/lending rates for loans maturing at $t_i$.

- In the case $K < (S_0 - PV(D))\, e^{rT}$ we borrowed $D_i$ at $t_i$, but since $t_i > 0$ is in the future we in principle should use a FRA to lock in the borrow rate to ensure there is no uncertainty in payoff at maturity

iii) case of stock with continuous dividend yield q:

Consider two portfolios at time t = 0:

portfolio A: same as above, one long forward $F_0$ + cash = $Ke^{-rT}$

portfolio B": $e^{-qT}$ unit of asset with all dividend reinvested in the asset.

At t = T, B" grows into one unite of asset, hence $A_T = B_T" = S_T$

→ at t = 0, $F_0 + Ke^{-rT} = S_0 e^{-qT}$ . $F_0 = 0$ → $K = S_0\, e^{(r-q)T}$

Arbitrage opportunities if $K \neq S_0\, e^{(r-q)T}$

- $K > S_0\, e^{(r-q)T}$ :

  at t = 0, short forward, borrow $S_0\, e^{-qT}$ to buy $e^{-qT}$ share of stock at $S_0$;

  at T, $e^{-qT}$ share of stock becomes 1 share, now priced at $S_T$, close forward with the share, receive K, use $S_0\, e^{(r-q)T}$ to repay bank,

  → pocket $K - S_0\, e^{(r-q)T}$

- $K < S_0\, e^{(r-q)T}$ :

  at t = 0, long forward, short $e^{-qT}$ share of stock at $S_0$, deposit the proceed to earn interest r;

  at T, close forward with K from the deposit (= $S_0\, e^{(r-q)T} > K$), use the stock got from the forward to close the short stock position (now due to reinvestment, it grows to 1 share)

  → pocket $S_0\, e^{(r-q)T} - K$

Note:

- In the above, if q is not constant but a known function of time $q_t$, then one should replace $e^{-qT} \to e^{-\int_0^T q_s ds}$ (exercise)

- Since in the continuous case all dividends are reinvested back into shares, the stock lenders do not require passing them back in the interim. In stead, for x share borrowed at t = 0 one needs to return $xe^{qT}$ shares at t = T.

iv) case of a forward contract f after inception and before maturity

Consider at time t ($0 < t < T$) one holds

portfolio A: a forward contract, $f_A$, to buy the underlying asset for K at time T, initiated at t = 0.

portfolio B: a forward contract, $f_B$, to sell the asset for F at time T, initiated at t = t. At t, $f_B = 0$.

- If S pays no dividend:

At time T, the payoff for A = $S_T - K$, for B = $F - S_T$, hence A + B = F – K. This is deterministic cash flow. Thus at t, $f_A + f_B = f_A = (F - K)\, e^{-r(T-t)}$

$F = S_t\, e^{r(T-t)} \rightarrow f_t = f_A = S_t - K\, e^{-r(T-t)}$

Arbitrage:

- $f_A > (F - K)\, e^{-r(T-t)}$ :

  at t, long forward deliver at F, short forward deliver at K.

  long = 0 → initial cash flow = $f_A$ (can be either positive or negative)

  at t = T, long + short = K – F

  → net risk free profit = $f_A\, e^{r(T-t)} - (F - K)$

- $f_A < (F - K)\, e^{-r(T-t)}$ :

  at t, long forward deliver at K, short forward deliver at F.

  short = 0 → initial cash flow = $-f_A$ (can be either positive or negative)

  at t = T, long + short = F – K

  → net risk free profit = $(F - K) - f_A\, e^{r(T-t)}$

Does not matter $f_A$ is positive or not.

- If S pays discrete dividend (exercise),

$F = S_t - PV_t(D) - K\, e^{-r(T-t)}$

- If S has continuous dividend yield q (exercise),

$f = S_t\, e^{-q(T-t)} - K\, e^{-r(T-t)}$

- Similarly for a forward contract to sell the underlying asset (exercise):

$f = (K - F)\, e^{-r(T-t)}$ (non-dividend paying stock), ….

v) case of a FRA after inception and before maturity

Consider at time t ($0 < t < T_1 < T_2$) one holds

portfolio A: a FRA, $f_A$, with notional N to receive rate $R_K$ for the period $[T_1 , T_2]$ , initiated at t = 0.

portfolio B: a FRA, $f_B$, with notional N to pay rate $R_F$ for the period $[T_1 , T_2]$, initiated at t = t. At t, $f_B = 0$.

The holder of A, B pays/receives realized reference floating rate $R_M$.

At time $T_2$ , the payoff for A = $N(R_K – R_M)(T_2 – T_1)$, B = $N(R_M – R_F)(T_2 – T_1)$, thus A + B = $N(R_K – R_F)(T_2 – T_1)$ , which is deterministic cash flow.

→ At t, $f_A + f_B = f_A = f = N(R_K - R_F)(T_1 - T_2)e^{-r_{t,T_2}(T_2-t)}$

Forward Foreign Exchange (FX) Rate & Forward Contracts on FX

Assume US dollar is the domestic currency

- Let $FX_0$, $FX_T$ = spot and forward FX rates at time T (1 foreign ccy = $FX);

  $r_d$ , $r_f$ = domestic and foreign interest rates

  Consider $PV of 1 foreign currency:

  $FX_0 e^{-r_f T} = e^{-r_d T} FX_T$ ("covered interest parity")

  Or equivalently, consider 1 foreign currency to be converted into $ at T. It can be exchanged either now or at T via forward:

  $FX_0 e^{r_d T} = e^{r_f T} FX_T$

  → $FX_T = FX_0 e^{(r_d - r_f)T}$

  → Similarity with forward price on a stock with continuous dividend yield

- Forward contract on FX:

  Long FX forward = receive pre-determined amount (contract size) foreign ccy, pay US dollar (using pre-determined (i.e., forward) FX rate)

  Forward points = forward rate – spot rate = $FX_T – FX_0$ (= "forward margins")

  Forward points > 0 : "premium"

  Forward points < 0 : "discount"

**Futures**

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Exchange traded forward contracts

- forwards: OTC between two counter parities (CP), custom made
  - counter party risk
  - finding counter parties may not be easy
  - early exit of a forward may be difficult
- futures: standardized contracts (expirations, sizes, delivery)
  - future exchanges: CME, NYSE-Euronext, Eurex, etc
  - future's underlying: commodity, currency, equity, equity indices, interest rate
  - commodities: where to deliver, quality specification
  - daily price movement limit
  - position limits
  - daily mark to market (MTM) & margin maintenance
    - initial margin
    - maintenance margin – margin call
    - margin requirement from exchanges vs from brokers
    - margin requirements for bona fide hedgers vs speculators (day traders vs long term position holders, netting)

Currently (2025) crypto futures markets account for about 75–80% of total cryptocurrency trading volume!

<u>Equity futures</u>

- expiries: March, June, September, December + next quarter, 3rd Friday of the month
- contract multiple
- for S&P 500 = $250 (value of a index point)

  S&P 500 E-mini size = $50, micro E-mini size = $5

Tax treatment on future trading

Qualified futures contracts traded in the US regulated exchanges (e.g., futures on regulated commodities, indices and certain cryptocurrencies) are subject to special tax treatment:

- 60/40 rule
- MTM accounting

  → may result in tax liability on unrealized gains
- Loss carry back (up to 3 years)
- Exempt from wash sale rule

→ Incentive for trading futures instead of cash

Arbitrages in future markets

"Cash-and-carry":

Recall for a non-dividend paying stock $F = S\, e^{rT} = S + S(e^{rT} - 1)$

$(e^{rT} - 1)$ = finance cost, part of "carry"

In general for an asset S,

$F = S(1 + C)$, C = carry

Carry charge for an asset may include (can be either positive or negative)

- finance
- payout from the asset
- insurance
- storage
- transportation

Financial asset:

- the latter 3 are negligible in general
- implied repo rate:

  Assuming non-dividend paying asset:

$F_{market} / S - 1 = C_{implied}$ → future market implied carry

→ future market implied repo rate

e.g., continuous compounded rate

$$r_{implied_repo} = \frac{1}{T}\log(1 + C_{implied})$$

Evaluate arbitrage opportunity by comparing future implied repo rate and repo market

Manufacture (replicate) a forward/future contract:

- Long: long asset funded by risk free borrow, carry (asset + loan) to maturity
- Short: short asset and deposit proceeding to earn risk free rate, carry (-asset + deposit) to maturity

Arbitrage:

- $F > S(1 + C)$: short future, manufacture a long forward position → risk free profit $= F - S(1 + C)$
- $F < S(1 + C)$: long future, manufacture a short forward position → risk free profit $= S(1 + C) - F$

The latter case is hard for non-financial asset (e.g., hard to borrow commodity asset). However the trade can still be carried out by a long term asset owner.

<u>Using futures to hedge price risk</u>

- long/short hedge
- cross hedge
- basis risk & optimal hedging ratio

  (more later in the course)

<u>Effects of future trading on the cash markets</u>

- "Flash Crash" in 2010

  Caused by the unusual activities in the S&P 500 future markets (?)

Misuse of futures - examples

- Nick Leeson's unauthorized and reckless trades on Nikkei 225 futures resulted in $1.3 bn loss and caused the demise of Barings Bank in 1995
- A large, misplaced bet on natural gas future term structure triggered $6.6 bn loss and the demise of Amaranth Advisors in 2006

**Arbitraging in Real World**

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Market Friction:

- transaction cost (both on futures and underlying assets): commission, bid/ask spread, market impact of the trade
- asset borrow cost b > 0 / restriction on short selling
- unequal borrow and lending rates, most likely $r_l < r_b$
- tax

Example – assume commission on S trading = $h_C$ percent:

  - cost of establishing a long position of S: $S_{ask}(1 + h_C)$
  - proceeding of short sell of S: $S_{bid}(1 - h_C)$

  → arbitrage boundaries for a forward on a non-dividend paying stock:

$$F_{bid} > S_{ask}(1 + h_C)\, e^{r_b T}, \quad F_{ask} < S_{bid}(1 - h_C)\, e^{(r_l - b)T}$$

Stock Index Arbitrage:

Exploit discrepancies between actual and theoretical futures prices.

Example:

Simultaneous buying/selling S&P 500 future while selling/buying the underlying stocks of S&P 500

→ underlying asset of the future is a (large) basket of stocks

→ executing through program trading

Practical issues:

- dividend estimation: harder for longer termed contracts
- index composition may change over time
- short order may take longer time to fill
- legging: future and spot position may not be established at the same time
- practice of index replication – potential tracking risk

Arbitrage in Treasury Markets:

Explore price discrepancies between the cash securities and the futures contracts in US treasury markets.

→ Treasury cash-futures basis trades

Popular among hedge funds

Example:

Long cash treasuries financed by repo, short futures

Practical issues:

- Margin on the future positions
- Fluctuations in repo rates
- Market liquidity

Arbitrageurs help keeping markets efficient!